

Mairin O'Shaughnessy

mairinoshaughnessy@gmail.com | 651-917-9151

Massachusetts Institute of Technology (MIT)

Candidate for BS in Computer Science and Molecular Biology

GPA: 4.9/5.0

Coursework includes: Intro to Probability, Software Construction, Linear Algebra & Optimization, Intro to Algorithms, Computation Structures, Differential Equations, Genetics, Organic Chemistry, Robotic Manipulation, Intro to Diffusion Models, Computational Proteomics, Intro to C and Assembly

Cambridge, MA
May 2027

Work Experiences

TriMin Systems

Roseville, MN

Software Engineering Intern

Jun 2025–Aug 2025, Jun 2024–Aug 2024

- Developed comprehensive and concise unit tests for legacy web application to integrate test-driven development into the workflow, utilizing JUnit 5 and Mockito frameworks and identifying bugs and areas for improvement
- Improved user experience by fixing UI issues, improving error handling, and patching security vulnerabilities for Spring application

Undergraduate Research Projects at MIT

Cambridge, MA

Semantic Segmentation for Graphite Phase Classification (Bazant Group)

Sep 2024–May 2025

- Created pipeline for automatically processing optical microscopy videos of graphite, minimizing time needed for pre-processing and supporting analysis of properties of graphite in lithium-ion batteries
- Implemented a neural network using PyTorch for colorimetric segmentation of graphite particles, overcoming class imbalance issues by utilizing weighted categorical loss and LAB color space

Computer Vision for Plant Density Analysis (Nepf Lab)

Jan 2024–May 2024

- Determined models for predicting stem count from images of marsh plants, utilizing the OpenCV library and computer vision methods to eliminate the need for hand counting stems when engineering sustainable marsh infrastructure
- Leveraged shape and color identification to design an efficient method for cropping test images, reducing time spent on processing experimental data

Self-Employed and MIT Eta Kappa Nu

Virtual and Cambridge, MA

Mathematics Tutor

Oct 2025–Present

- Tutoring high school and college students in various math classes, including Advanced Geometry and Introduction to Probability

Selected MIT Course Projects

Creating Video Games - Hush

Fall 2025

- Collaborated with a team of six students to design and create a medieval themed side-scrolling video game using an Agile framework, working within the theme of "Playful Audio"
- Designed and integrated animation assets for the game using Unity Engine and C#

Robotic Manipulation - StackBot

Fall 2025

- Implemented the perception module for a full-stack system for manipulating a virtual 6-DoF robotic arm to stack blocks of varying sizes
- Utilized the Drake robotics library to crop and filter 3D environment points and find the positions of objects using geometric pose estimation, creating further geometric techniques to estimate object sizes

Awards and Honors

FIRST Dean's List Finalist Award

Apr 2022

- One of eight Minnesota students recognized for leadership and technical expertise on a FIRST Robotics Competition Team
- Acknowledged for my effectiveness as team captain, mentorship of younger students from elementary to high school, and technical expertise in robotic programming in Java

Eta Kappa Nu Honor Society

Oct 2025

- Invited to join the international honor society for computer science students in recognition of exceptional completion of coursework